

SUMMARY

AI Engineer with hands-on experience developing and operationalizing end-to-end AI solutions across classical ML, deep learning, and generative AI, covering everything from data preparation to production deployment of RAG pipelines, LLM-powered applications, and model inference systems.

TECHNICAL SKILLS

**Programming:** Python, SQL, C/C++, Java, JavaScript, Bash  
**ML & AI:** PyTorch, Scikit-learn, Pandas, NumPy, OpenCV  
**GenAI & LLMs:** RAG, Vector DBs, Prompt Engineering  
**Databases:** MySQL, PostgreSQL, SQLite, MongoDB  
**DevOps & Tools:** Docker, FastAPI, REST APIs, Git, CI/CD

EXPERIENCE

**AI Research Assistant** Sep. 2025 – Present  
*Eötvös Loránd University (ELTE)* Budapest, Hungary

- Build and optimize DP-SGD federated learning pipelines for privacy-preserving synthetic data generation ( $\epsilon = 1$ ), engineering data preparation and feature workflows to benchmark privacy-utility trade-offs
- Author and teach the open-source IDaSec Practicum (8 labs), documenting AI architectures and best practices in responsible AI
- Collaborate with cross-functional teams and mentor graduate students in AI engineering and MLOps workflows

**Data Analyst** Dec. 2024 – Mar. 2025  
*Djezzy* Khenchela, Algeria

- Analyzed subscriber telecom datasets using Python and SQL to identify usage trends and customer behavior patterns
- Built automated data preparation and feature engineering pipelines, facilitating data-driven decision making and reporting
- Collaborated with analytics and cross-functional teams to model subscriber metrics and optimize customer retention strategies

PROJECTS

**CASPER** 2026  
*Python, Differential Privacy, RAG, T5, MedCPT, Docker*

- Built a privacy-preserving RAG framework that clusters sensitive medical records with DP-KMeans and inverts noisy centroids into synthetic prototypes, enabling LLMs to retrieve on private data without accessing raw records
- Achieved technical Recall@10 of 0.91 at  $\epsilon \approx 44.28$ ,  $\delta = 10^{-5}$ , with 50% MIA AUC (random-guessing), demonstrating formal and empirical privacy guarantees

**Context-Aware Access Control for IoT Device** 2026  
*FastAPI, Python, Multi-Agent LLMs, Docker, Wazuh, Prometheus* ITAT'2026, WAS

- Built a hybrid zero-trust IoT authorization framework combining a deterministic validation pre-filter ( $< 1$  ms fast path) with multi-agent LLM reasoning (97 ms slow path) for autonomous context-aware access control
- Achieved 94.1% decision accuracy vs. 82.4% for the RBAC baseline and blocked 100% of 105 red-team adversarial attacks, with fail-secure DENY defaults on LLM rate limits or timeouts

**MyVellum** Dec. 2025 – Present  
*FastAPI, Groq API, LLM Orchestration, RAG, Docker, CI/CD*

- Architected an LLM-powered app with modular prompt engineering, vector-backed RAG pipelines, and retrieval optimization
- Built automated evaluation gates into CI/CD for relevance and quality metrics; integrated Groq API and FastAPI with Docker

EDUCATION

**PhD in Informatics** (GPA: 5/5) Sep. 2025 – Expected 2029  
*Eötvös Loránd University (ELTE)* Budapest, Hungary  
*Thesis: Applied synthetic data generation for secure, privacy-preserving AI systems (Supervised by Pr. Lendák Imre)*

**M.Sc. in Artificial Intelligence – Ranked 1st in Class** (GPA: 3.7/4.0) Sep. 2023 – Jun. 2025  
*University of Khenchela* Khenchela, Algeria  
*Thesis: Hair Dermoscopic Image Segmentation Using Deep Learning (Supervised by Pr. Dalal Bardou)*

**B.Sc. in Mathematics and Computer Science – Ranked 1st in Class** (GPA: 3.9/4.0) Sep. 2020 – Jun. 2023  
*University of Khenchela* Khenchela, Algeria  
*Final Year Project: A learning management system for Algerian students.* 

AWARDS & CERTIFICATIONS

- **Stipendium Hungaricum Scholarship:** Fully funded PhD studies at ELTE Sep. 2025
- **2nd Place — Starthon'25 Hackathon:** University of El Oued, Algeria Apr. 2025
- **Software Engineering (Back-end Specialization):** ALX Africa ([savanna.alxafrica.com/certificates/yNTLX6BYHR](https://savanna.alxafrica.com/certificates/yNTLX6BYHR)) Feb. 2025
- **AI Career Essentials (AiCE):** ALX Africa ([savanna.alxafrica.com/certificates/YEM9LC5BNm](https://savanna.alxafrica.com/certificates/YEM9LC5BNm)) Jul. 2024